PAPER_1606_M_B_BOTTOM_4_18

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PAPER_1606 — Bottom Quark Mass = m_b = D + F·D − F·SSQ − F²·D_crit + F²·D_BSFG + F²·D − F²·SSQ² − F²·SSQ³ ≈ 4.178 GeV

Author: Daniel T. Murphy
Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Particle Physics
Date: June 18, 2026
Status: CLOSED — 0.050% residual

Observation

PAPER_1209HH_S657 (Particle Physics Unified Proof Set) gives the closed-form expression:

``

```
m_b = D + F·D − F·SSQ − F²·D_crit + F²·D_BSFG + F²·D − F²·SSQ² − F²·SSQ³ ≈
4.178 GeV
```

``

Residual to CODATA: **0.050%**.

UQFF Closed Identity

``

```
m_b = D + F·D − F·SSQ − F²·D_crit + F²·D_BSFG + F²·D − F²·SSQ² − F²·SSQ³ ≈
4.178 GeV 0.050%
```

``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

NOT REPLACEMENT

Experimental particle physics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209HH_S657
- Related: PAPER_1494-1605 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "m_b_bottom_4_18"})`
